Peixing You

peyou@ucsd.edu | github.com/ypx19 | Google Scholar

Education

University of California, San Diego San Diego, CA
MS in Robotics, Department of Electrical and Computer Engineering *Sep. 2025 – Present*

- GPA: 3.8/4.0
- Advisor: Prof. Xiaolong Wang

Tsinghua University Beijing, China
BS in Materials Science and Engineering; BS in Computer Science *Aug. 2019 – Jun. 2024*

- Advanced GPA, last three years: 3.58/4.0
- Advisors: Prof. Yang Liu, Prof. Peng Li, Prof. Huaping Liu

Selected Experience

Optimization-Driven Robotic Hand Design UC San Diego, San Diego, CA
Robotics Research Prototype *Sep. 2025 – Dec. 2025*

- Developed a neural optimization pipeline with PyTorch and NVIDIA Warp, using forward kinematics and human finger-motion data to explore **low-cost dexterous hand designs** better suited for real-world manipulation tasks and teleoperation.
- Converted optimization outputs into printable geometry via mesh insertion and intersection processing, then fabricated the robotic hand with 3D printing and tested its **real-world collision** behavior to evaluate feasibility for future manipulation tasks.
- Mitigated the **Sim2Real gap** by improving simulation fidelity and refining the optimization objective with a more accurate Signed Distance Field-based collision metric, reducing inter-finger collision cases in the physical prototype by **50%**.

Interactive Robotic Grasping System Tsinghua University, Beijing, China
Robotics System Project *Jun. 2023 – Sep. 2023*

- Built an **IEEE CYBER 2024 Best Poster** Award-winning closed-loop robotic grasping pipeline with deep neural networks and ROS, using RGB-D camera input and point-cloud reasoning to retrieve target books from shelf environments and demonstrate a reliable approach to object retrieval in crowded scenarios such as markets, libraries, and warehouses.
- Integrated perception, robotic arm control and policy execution modules by multi-node ROS architecture for implementing the algorithm design.
- Implemented long-horizon target memory using IoU and MSE metrics to match observations over time, allowing the system to preserve target identity throughout multi-step interactions.
- Configured and tested a Linux-based hardware platform to access GPU computing resources for real-world algorithm experiments, gaining hands-on experience in hardware setup, camera input, robot control, and validation workflows.

Active Perception Policy for Robot AIR, Tsinghua University, Beijing, China
Research Assistant *Jun. 2024 – Dec. 2024*

- Identified active perception as a valuable direction for robotic perception through literature review, prototype implementation, and feasibility testing, highlighting its potential to improve environmental understanding and support real-world tasks such as navigation and object localization.
- Designed and evaluated an LLM-based active-perception policy with a memory stack for indoor robotic object localization, achieving nearly 3× higher efficiency than the strongest baseline (**17 vs. 6 SPL**), and identifying both the task categories that benefited from active perception and the limitations of current LLM reasoning and modeling.

Shape-Aware 3D Keypoint Neural Network AIR, Tsinghua University, Beijing, China
Research Assistant *Oct. 2020 – Jun. 2021*

- Developed SNAKE, a **NeurIPS 2022 Spotlight** paper, a shape-aware self-supervised neural 3D keypoint field that jointly predicted surface occupancy and keypoint saliency from continuous 3D queries, coupling shape reconstruction with repeatability, surface-consistency, and sparsity objectives to achieve up to **85%** repeatability and strong robustness to down-sampling and noise.
- Rapidly reproduced and extended PyTorch-based 3D keypoint detection baselines for following method study and comparison experiments.
- Optimized Python experiment scripts using NumPy vectorization to improve efficiency across repeated model evaluation

workflows.

Gello Teleoperation System Implementation UC San Diego, San Diego, CA
Robotics Research Assistant *May. 2026 – Jun. 2026*

- Built a Gello-based **teleoperation platform** by integrating a YAM passive robotic arm, Dynamixel actuators, and control software for robot **manipulation data collection**.
- Diagnosed and resolved cross-layer integration issues spanning **CAD design**, hardware assembly, motor communication, and software configuration during system deployment.
- Improved the open-source Gello ecosystem by identifying incorrect component specifications and missing 3D models, submitting validated fixes through a GitHub pull request.

Self-Balancing Robot from Scratch Xining, China
Personal Project *May. 2025 – Jul. 2025*

- Built a **real-time embedded control system** for a self-balancing robot using Arduino Nano, MPU6050 IMU, encoder-equipped motors, and an HC08 motor driver; implemented interrupt service routines (ISRs) for encoder feedback, IMU sampling, and low-latency motor control.
- Designed a **dual-loop PID controller** that regulates the robot's pitch angle through nested angle and velocity feedback, eliminating long-term drift observed with single-loop balancing and improving recovery robustness.
- Implemented interrupt-driven encoder processing with both fixed-time and fixed-pulse **velocity estimation**, evaluating their responsiveness and noise characteristics to achieve reliable motor feedback for real-time control.

Experience

Agent Arena - AI Agent Simulator Cambridge, MA
Backend / AI Engine Startup Project *Mar. 2026 – Present*

- Translated user requirements into an **LLM-powered simulation system** by designing and implementing backend features with Python and FastAPI. Contributed to the company's acquisition of **real paid startup customers**.
- Integrated MinIO object storage after rapidly learning the related infrastructure, enabling **scalable multi-file handling** that is essential for user interaction with the LLM simulation system.
- Diagnosed long-context blocking failures in LLM workflows and implemented timeout-handling logic, reducing worst-case pipeline blocking time by **96.7% from 300s to 10s** and improving **reliability and development efficiency**.

Baidu – LLM Spreadsheet Processing Agent Beijing, China
LLM Agent Engineering / Testing Intern *Apr. 2024 – Jun. 2024*

- Benchmarked different candidate solutions for an LLM-powered spreadsheet processing agent by building evaluation workflows, experiment scripts, and result-checking logic to assess response reliability, format consistency, and downstream compatibility, advancing R&D progress and supporting product development decisions.
- Implemented multi-threaded evaluation workflows and structured test datasets with Python to benchmark spreadsheet-processing outputs at scale, reducing repetitive manual verification and accelerating experimentation.

LLM Interpretability and Behavior Alignment Columbia University & Tsinghua University
Research Assistant *Apr. 2024 – May. 2024*

- Developed an experimental pipeline to interpret LLM internal representations by extracting and manipulating **intermediate-layer token activations** from LLaMA-based models.
- Conducted causal intervention experiments on transformer activations to investigate how layer-wise representations influence model behaviors and generated responses.
- Explored token-level interpretation guided **parameter-efficient fine-tuning** methods for LLM knowledge and behavior editing, identifying behavior-relevant representations to enable resource-efficient targeted modification, including factual updates and safety-oriented semantic steering.

AI Hardware Prototype Advx25 Hackathon, Hangzhou, China
Hackathon Project *Jul. 2025 – Aug. 2025*

- Built a vision-enabled AI earphone prototype within three days by integrating an ESP32 camera module, Bluetooth communication, speaker output, mobile-side triggering, and LLM connectivity, enabling users to capture what they see and receive real-time conversational assistance.
- Led **hardware-software integration** for the end-to-end interaction pipeline, including ring-based wake-up, camera capture, Bluetooth data transfer, and phone-side processing; evaluated BLE vs. classic Bluetooth trade-offs and used chunked transmission to work around BLE packet-size limitations.
- Refined the product from a broad AI-earphone interaction concept into concrete user scenarios and technical requirements through repeated discussions with investors, teammates, and technical mentors, narrowing use cases toward

visually impaired users and context-heavy business scenarios while identifying key constraints such as latency, battery life, camera occlusion, stabilization, and wake-up mechanism design.

Object-Oriented Game Simulation Tsinghua University, Beijing, China
Course Project *Mar. 2023 – Jun. 2023*

- Translated game design and functionality requirements into a modular C++ framework, defining classes and methods for game entities, state management, and interaction logic.
- Implemented dynamic scene transitions and reversible game-state updates using stack-based data structures.
- Implemented consistent state transitions across player actions and scene updates by designing event-driven game logic.

Skills

Languages: Python, C++, TypeScript, Java, SQL

Fronted/Backend Engineering: React, REST APIs, Backend API Design, Request/Response Handling, Authentication, Database, Object Storage, Error Handling, Logging, Timeout Handling

Infrastructure & Tools: Linux, Git, Docker, AWS, Mujoco, IssacGym, ROS, Pytorch, Warp

AI: LLM Agents, LLM Parameter-efficient Fine Tuning, Machine Learning, Reinforcement Learning, Representation Learning, Imitation Learning

Robotics: Robotic Control, PID, LQR, Manipulation, Navigation, Dexterous, Perception, Diffusion Policy, VLA

Embedded System: Interruption, Hardware-Software Integration, Embedded System Design, Embedded C, Bluetooth Communication, PWM Motor Control, IMU Estimation

CS Fundamentals: Object-Oriented Programming, Data Structures, Event-Driven Logic, Modular Design

Publications

1. Peixing You, Hang Li, et al. (2024). "An Adapter for Interactive Object Retrieval on the Shelf". In: *2024 IEEE 14th International Conference on CYBER Technology in Automation, Control, and Intelligent Systems (CYBER)*, pp. 505–510. DOI: `10.1109/CYBER63482.2024.10749016`. URL: `https://ieeexplore.ieee.org/document/10749016/`
2. Peixing You, Chengliang Zhong, et al. (2022). "SNAKE: Shape-aware Neural 3D Keypoint Field". In: *Advances in Neural Information Processing Systems*. Ed. by S. Koyejo et al. Vol. 35. Curran Associates, Inc., pp. 7052–7064. URL: `https://proceedings.neurips.cc/paper_files/paper/2022/file/2e3eccb54649186564ad6627ed80848c-Paper-Conference.pdf`

Awards and Honors

- 09/2024 – IEEE CYBER 2024 **Best** Poster Award, "An Adapter for Interactive Object Retrieval on the Shelf"
- 12/2022 – NeurIPS 2022 **Spotlight** Paper, "SNAKE: Shape-aware Neural 3D Keypoint Field"
- 03/2023 – Tsinghua Science and Technology Innovation Excellence Scholarship (**Top 1%**)
- 03/2023 – Tsinghua Learning Progress Scholarship (**Top 1%**)
- 05/2022 – Tsinghua MaYueHan Soccer Cup Third Prize
- 12/2020 – Chinese College Student Mathematics Competition Third Prize
- 05/2020 – Tsinghua Siege Robotic Competition Third Prize